

Current, Pressure and What a Breaker Actually Protects

The four quantities, and the one rule that explains half the defect questions on this topic.

Electricity is the flow of electrons along a conductor. Everything else in this topic is either a way of carrying that flow or a way of stopping it getting too big. Start by keeping the four quantities apart, because the exam writes questions that only work if you confuse them.

Voltage	The pressure pushing the flow. Measured in volts.
Current	The flow itself. Measured in amps. Also called amperage — when a question asks for the amount of electricity flowing through a wire, the word it wants is amperage.
Resistance	What opposes the flow. Measured in ohms.
Power	The work done. Measured in watts.
Conductor	The wire that carries the current. When a question says "electricity moves along wires known as", the answer is conductors, not cables and not circuits.

Volts push, amps flow, ohms resist, watts work. Say it in that order and the four never swap places.

The rule the whole topic rests on

A circuit breaker protects against too much current, and the maximum breaker rating must not exceed the allowed ampacity of the conductor. The breaker protects the wire, not the appliance. A wire has a fixed capacity to carry current without overheating, and the breaker exists so that capacity is never passed.

The numbers here

14 AWG copper — maximum overcurrent device	15 amp
12 AWG copper — maximum overcurrent device	20 amp
10 AWG copper — maximum overcurrent device	30 amp

AWG runs backwards: the smaller the number, the thicker the wire. Those three pairs are the whole basis of the classic defect question.

YOU HAVE MISSED THIS 2 TIMES

A 14-gauge copper wire is protected by a 20-amp fuse. You should report that:

This is a defect and recommended improvement.

The reason is mechanical: 14 AWG is rated to 15 amps, so a 20-amp fuse lets the wire carry 20 amps and overheat before anything trips. The protection is sized to the weakest thing it protects.

Say "maximum overcurrent protection", not "the wire is rated for"

NEC 240.4(D), the small conductor rule, caps the size of the protective device. A 10 AWG conductor may physically be rated above 30 amps, and the breaker still may not exceed 30. The exam phrases it as the device limit, so learn it that way.

Breakers, fuses and a hazard worth recognizing

Circuit breaker	Trips and switches off the circuit when current passes the system's capacity. Resets.
Fuse	Does the same job once, then must be replaced.
Fuse on both hot and neutral	A safety hazard that needs to be corrected. A blown neutral fuse leaves the circuit live but apparently dead.
Loose connection	A high-resistance point, and resistance makes heat. This is why loose terminals show up as burnt or melted insulation.

Say it back

Cover the page. Answer each one out loud.

1. Name the four quantities and their units, in the push-flow-resist-work order.
2. What does a breaker protect, and what may its rating never exceed?
3. Give the three AWG and amp pairs.
4. 14-gauge copper on a 20-amp fuse. What do you write, and why?
5. What is the word for the amount of electricity flowing through a wire?
6. Why is a fuse on the neutral a hazard rather than just unusual?